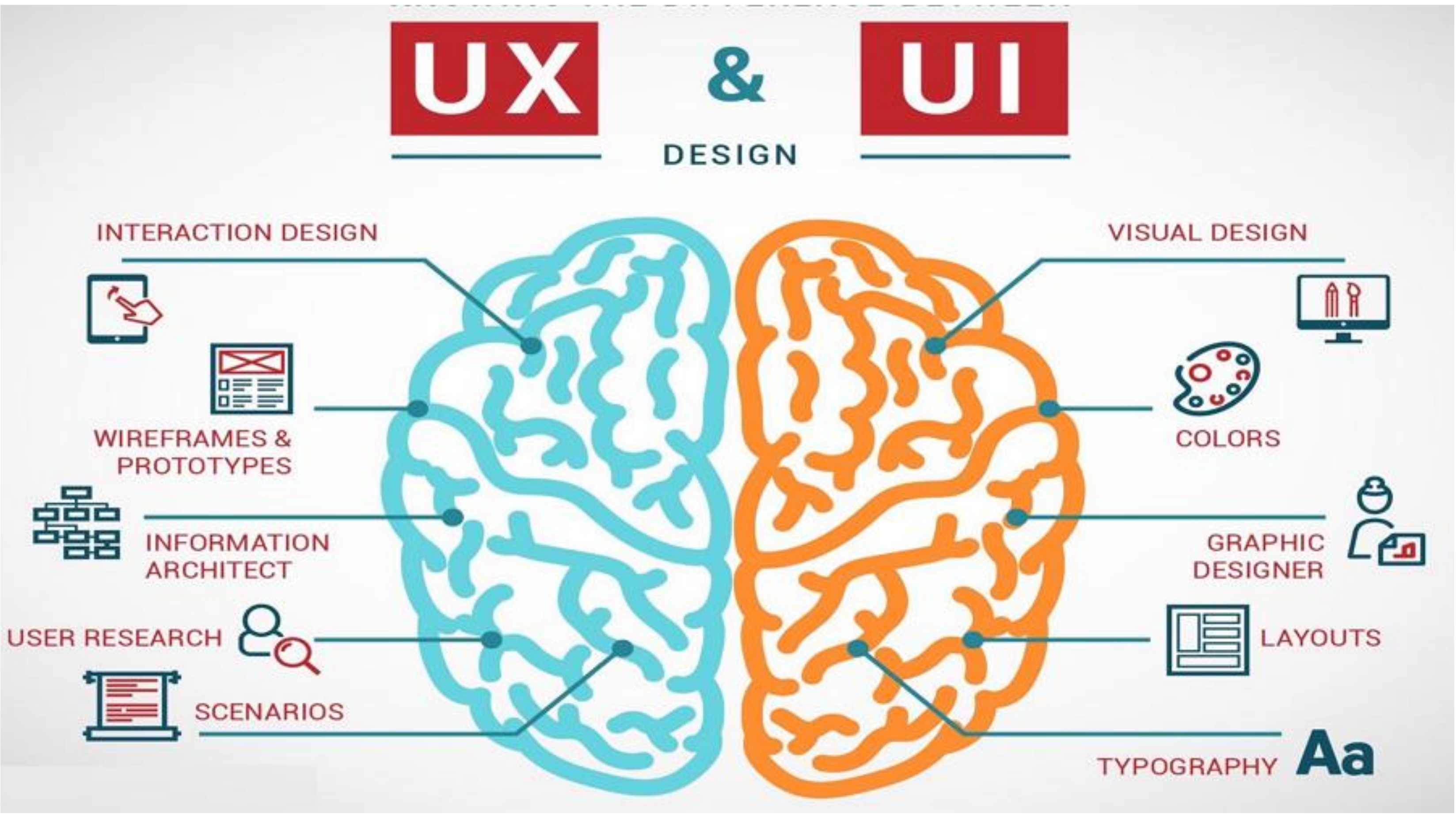
UI/UX Design

Lecture 6

Outline

- The difference between UX design and UI design
- Laws of UX:
  - Jacob's Law
  - Fitt's Law
  - Hick's Law
  - Miller's Law
  - Postel's Law and more in Jon Yablonski
- UI patterns:
  - Hierarchy
  - Gestalt principles
- Classroom Activity (Figma):
  - Icons, overlays, slicing, etc
  - Prototyping

UX design vs UI design



User Experience Design

- UX design is studying user behaviour and understanding user motivations with the goal of designing better interaction with a product or a software.
- Key topics in UX design:
    - Usability
    - Utility
    - Accessibility
    - Performance
    - Ergonomics
    - Design and aesthetics

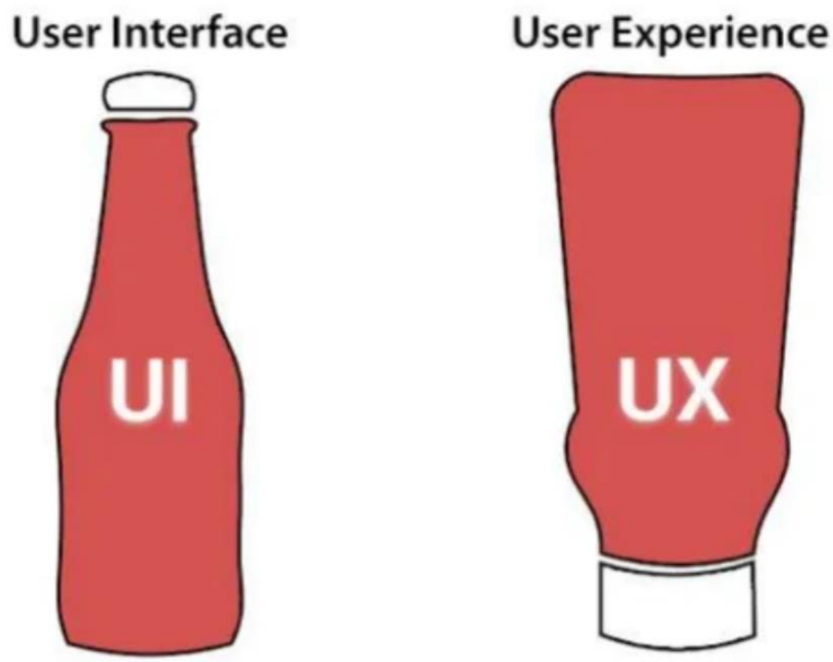
User Interface Design

- UI design is studying visual, voice, gesture or other perceptible elements that enable effective and pleasant interaction between users and a product
- Key topics in UI design:
  - Visual design elements:
    - lines, shapes, colours, texture, typography, forms
  - Relationship between visual elements:
    - layout, hierarchy, contrast, balance, scale, similarity, gestalt, whitespace
  - User controls:
    - textboxes, checkboxes, buttons, lists, grids, tables, accordions..

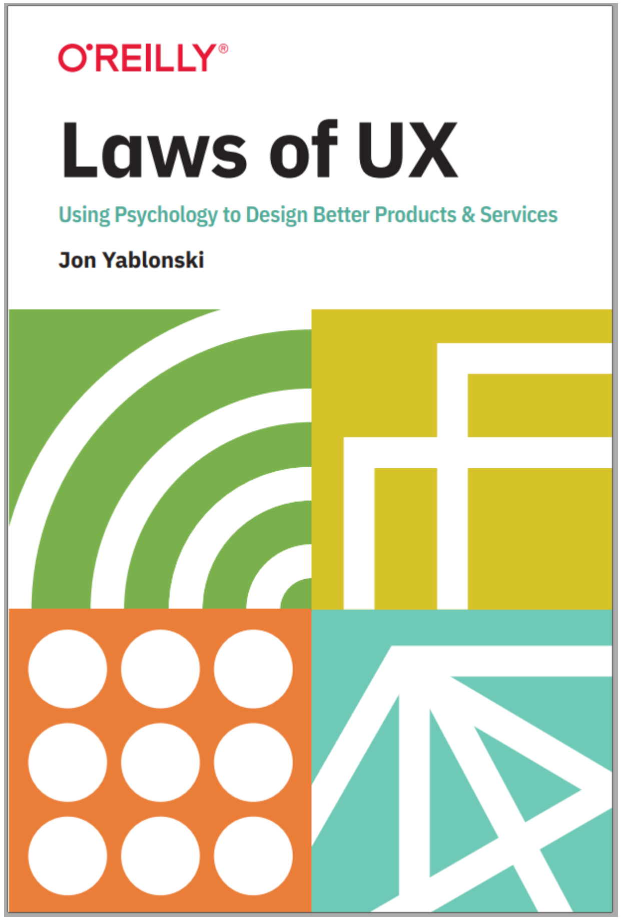
To sum up…

UI vs UX

- UX design is what makes an interface useful
- UI design is what makes an interface beautiful.



Laws of UX



- Contains more than 10 different principles that are drawn from psychology.
- Provides quick reference to well-known UX patterns

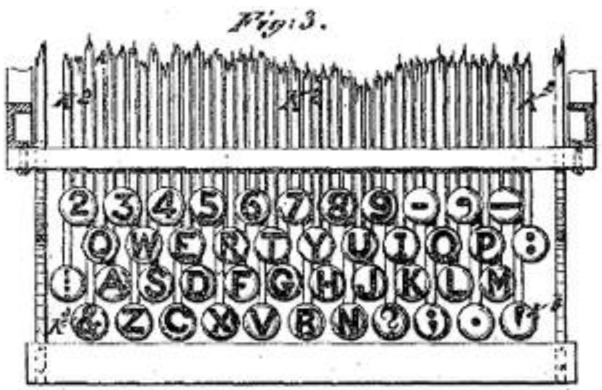
Jacob's Law

Users spend most of their time on other sites, and they prefer your site to work the same way as all the other sites they already know.

- Users will transfer expectations they have built around one familiar product to another that appears similar.
- By leveraging existing mental models, we can create superior user experiences in which the users can focus on their tasks rather than on learning new models
- When making changes, minimize discord by empowering users to continue using a familiar version for a limited time.

Jacob's Law Example

Christopher Latham Sholes's
1878 QWERTY keyboard layout



Modern QWERTY keyboard

Old typewriter QWERTY keyboard layout

iPhone Keyboard

Click Evolution breaks Jacob's Law

Jacob's Law Example

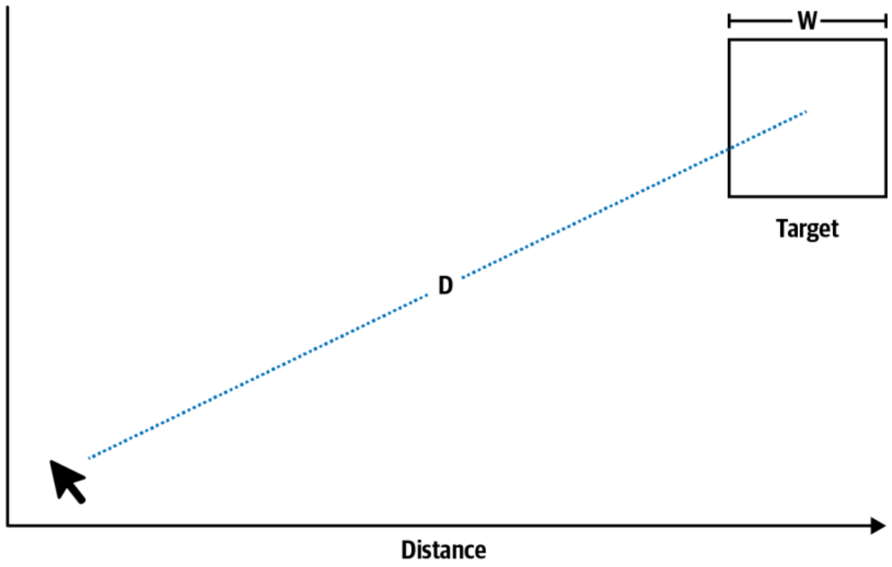
Figure 1-4. Seat controls in the 2020 Mercedes-Benz EQC 400 Prototype, informed by the mental model of a car seat (source: MotorTrend, 2018)

Fitt's Law

The time to acquire a target is a function of the distance to and size of the target.

- Touch targets should be large enough for users to accurately select them.
- Touch targets should have ample spacing between them.
- Touch targets should be placed in areas of an interface that allow them to be easily acquired.

Fitt's Law



Index of Difficulty

$$ID = \log_2\left(\frac{2D}{W}\right)$$

- Minimum Touch size 16mm x 16mms
- Minimum whitespace between elements 10mms-14 mms

Fitt's Law Examples

Choose emotions or stamps on spot by pressing E in FigJam

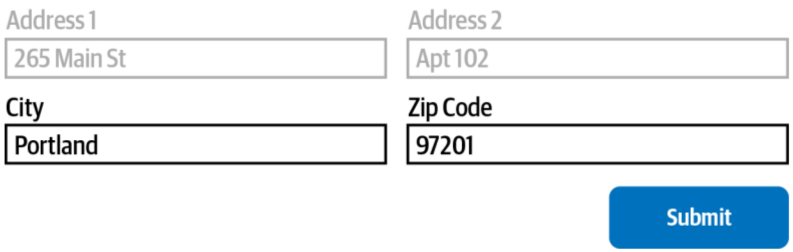


Figure 2-4. Form submission buttons are placed in close proximity to the last form input

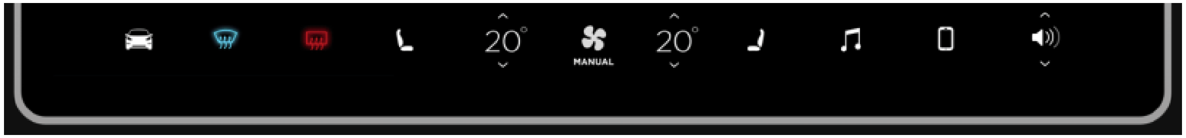


Figure 2-6. Providing sufficient space between items increases usability, minimizing the chances of selecting the wrong action (source: Tesla, 2019)

Fitt's Law Examples

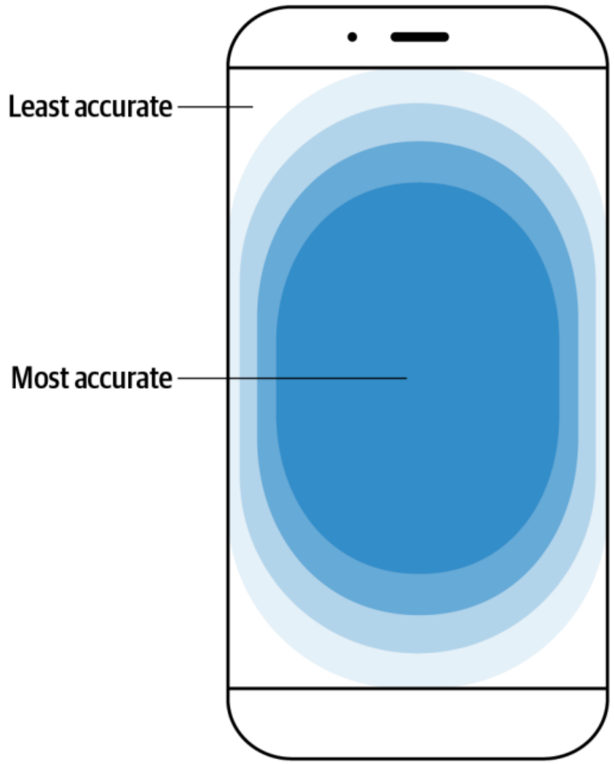


Figure 2-2. Smartphone touch accuracy (illustration based on research by Steven Hoober)

Figure 2-7. The iPhone's Reachability feature enables easy access to the top half of the screen (source: Apple, 2019)

Hick's Law

The time it takes to make a decision increases with the number and complexity of choices available.

- Minimize choices when response times are critical to increase decision time.
- Break complex tasks into smaller steps in order to decrease cognitive load.
- Avoid overwhelming users by highlighting recommended options.
- Use progressive onboarding to minimize cognitive load for new users.
- Be careful not to simplify to the point of abstraction.

Hick's Law

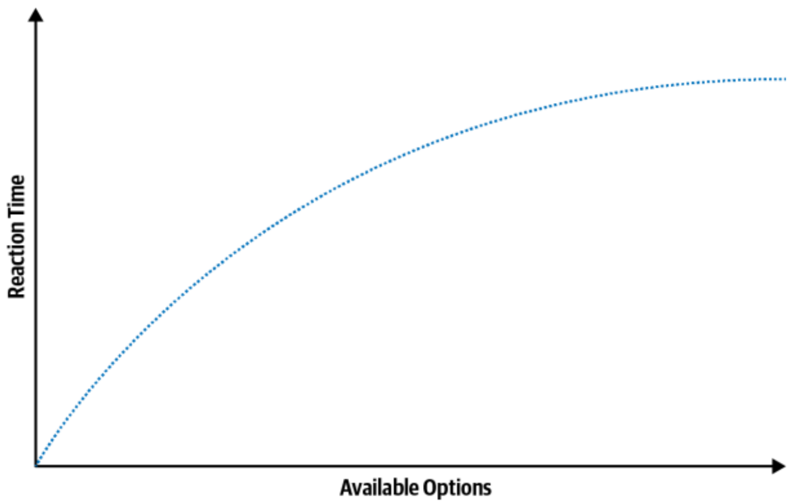


Figure 3-1. Diagram representing Hick's law

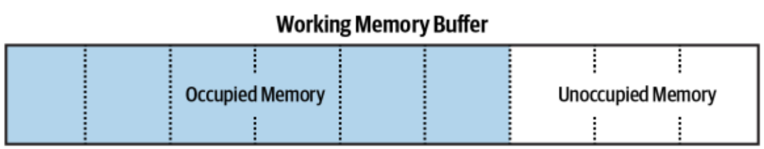


Figure 3-2. Working memory buffer illustration

Hick's Law Examples

Figure 3-3. Modified TV remotes that simplify the "interface" (sources: Sam Weller via Twitter, 2015 [left]; Luke Hannon via Twitter, 2016 [right])

Figure 3-4. A smart TV remote, which simplifies the controls to only those absolutely necessary (source: Digital Trends, 2018)

Hick's Law Examples

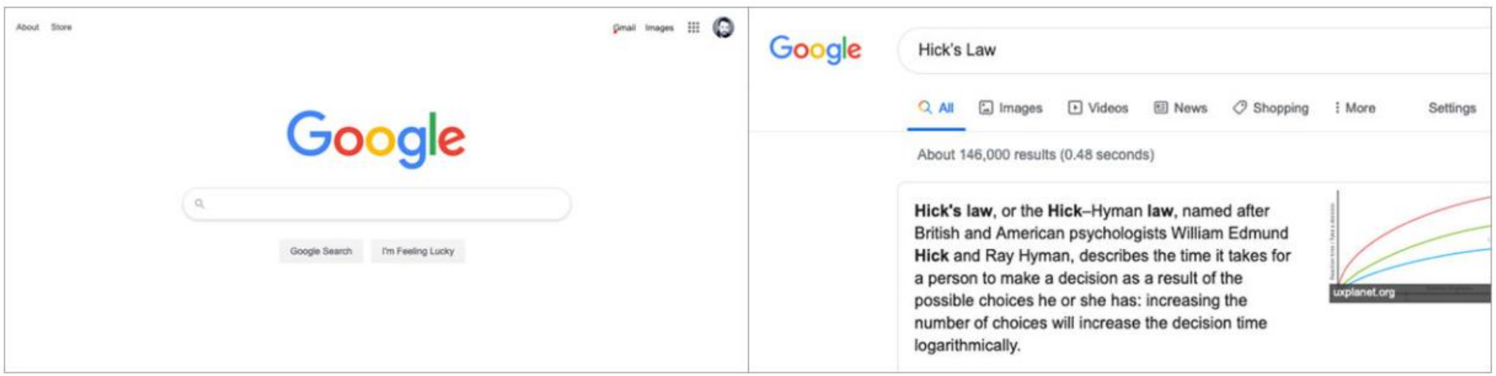


Figure 3-5. Google simplifies the initial task of searching (left) and provides the ability to filter results only after the search has begun (right) (source: Google, 2020)

Miller's Law

The average person can keep only 7 (± 2) items in their working memory.

- Don't use the "magical number seven" to justify unnecessary design limitations.
- Organize content into smaller chunks to help users process, understand, and memorize easily.
- Remember that short-term memory capacity will vary per individual, based on their prior knowledge and situational context.

Miller's Law

- Chunking - breaking down information into a number of information chunks for memorization.

Figure 4-1. A (US) phone number with and without chunking applied

Miller's Law Examples

A wall of text is an excessively long post to a noticeboard or talk page discussion, which can often be so long that some don't read it. Some walls of text are intentionally disruptive, such as when an editor attempts to overwhelm a discussion with a mass of irrelevant kilobytes. Other walls are due to lack of awareness of good practices, such as when an editor tries to cram every one of their cogent points into a single comprehensive response that is roughly the length of a short novel. Not all long posts are walls of text; some can be nuanced and thoughtful. Just remember: the longer it is, the less of it people will read. The chunk-o'text defense (COTD) is an alleged wikilawyering strategy whereby an editor accused of wrongdoing defends their actions with a giant chunk of text that contains so many diffs, assertions, examples, and allegations as to be virtually unanswerable. However, an equal-but-opposite questionable strategy is dismissal of legitimate evidence and valid rationales with a claim of "text-walling" or "TL;DR". Not every matter can be addressed with a one-

Figure 4-2. "Wall of text" example (source: Wikipedia, 2019)

Wall of Text

A wall of text is an excessively long post to a noticeboard or talk page discussion, which can often be so long that some don't read it.

Types

Some walls of text are intentionally disruptive, such as when an editor attempts to overwhelm a discussion with a mass of irrelevant kilobytes. Other walls are due to lack of awareness of good practices, such as when an editor tries to cram every one of their cogent points into a single comprehensive response that is roughly the length of a short novel. Not all long posts are walls of text; some can be nuanced and thoughtful.

Just remember: the longer it is, the less of it people will read.

Figure 4-3. "Wall of text" improved with hierarchy, formatting, and appropriate line lengths (source: Wikipedia, 2019)

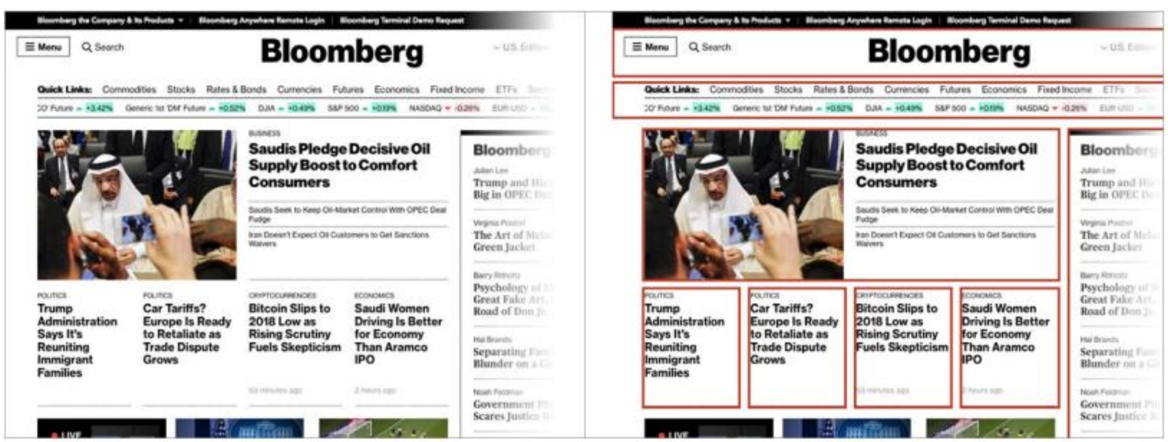


Figure 4-4. Example of chunking applied to dense information (source: Bloomberg, 2018)

Peak-End Rule

People judge an experience largely based on how they felt at its peak and at its end, rather than on the total sum or average of every moment of the experience.

- Pay close attention to the most intense points and the final moments (the “end”) of the user journey.
- Identify the moments when your product is most helpful, valuable, or entertaining and design to delight the end user.
- Remember that people recall negative experiences more vividly than positive ones.

Peak-End Rule

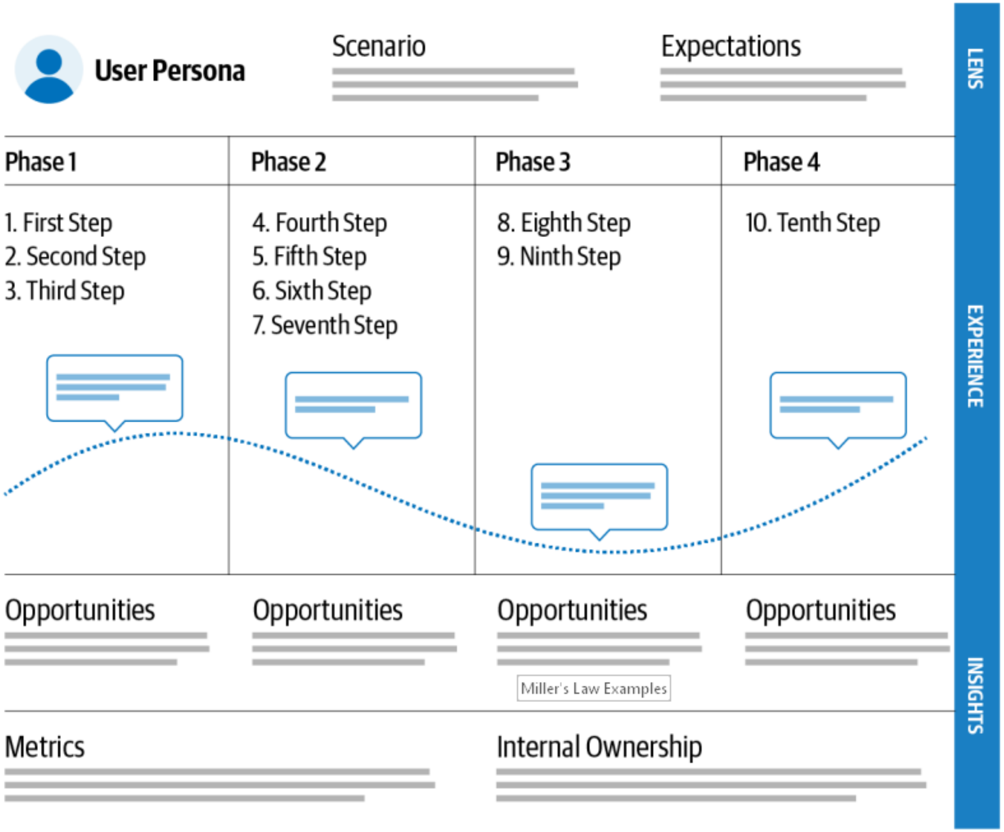


Figure 6-4. Example journey map

Peak-End Rule Examples

Peak

End

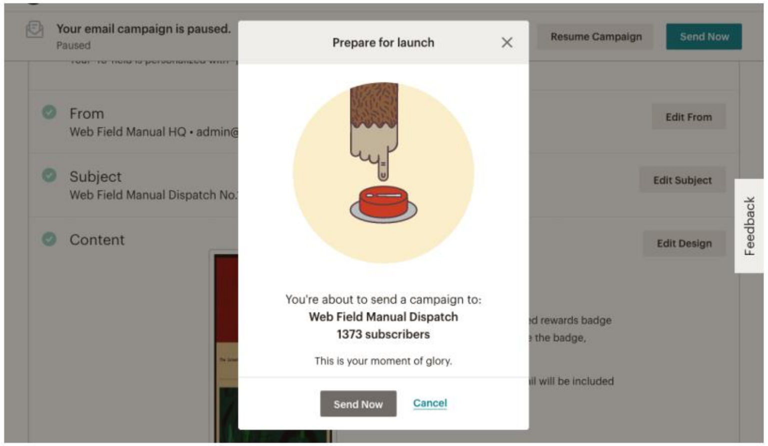


Figure 6-1. Mailchimp's email campaign confirmation modal (source: Mailchimp, 2019)

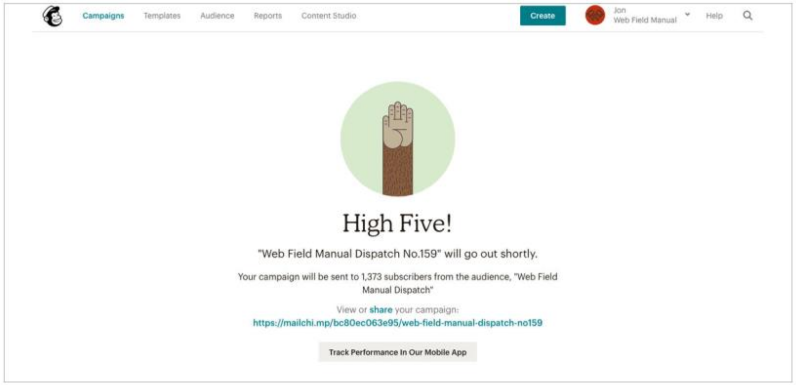


Figure 6-2. Mailchimp's "email sent" screen (source: Mailchimp, 2019)

Peak-End Rule Examples

Waiting is unavoidable and unpleasant experience. Uber applied following UX techniques to address this issue:

- Idleness aversion
- Operational transparency
- Goal gradient effect

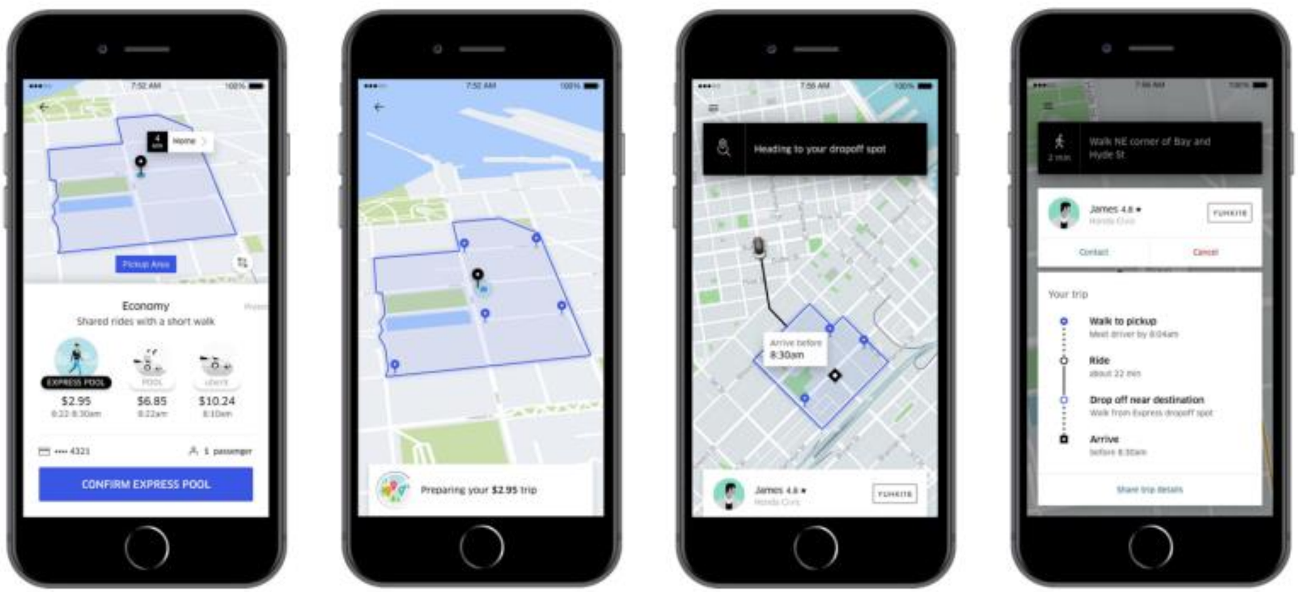


Figure 6-3. Uber Express POOL (source: Uber, 2019)

Peak-End Rule Examples

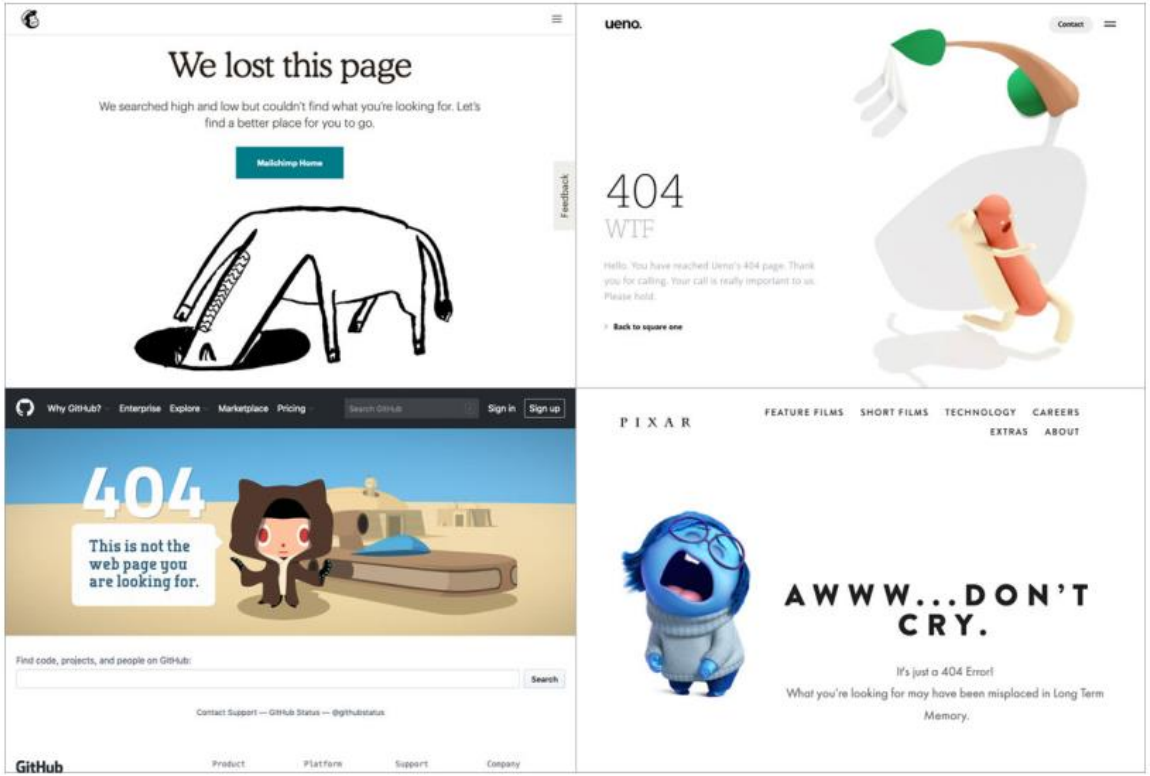


Figure 6-5. Various 404 pages that use humor and brand personality (sources [clockwise from top left]: Mailchimp, Ueno, Pixar, and GitHub, 2019)

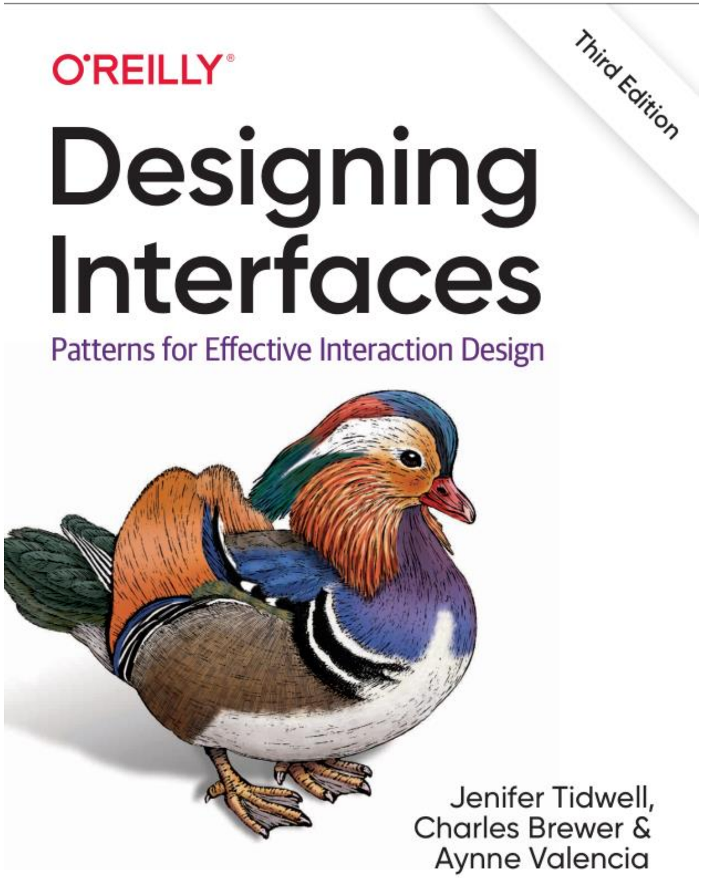
More UX Laws:

- Postel's Law
  - Be conservative in what you do, be liberal in what you accept from others.
- Aesthetic-Usability
  - Users often perceive aesthetically pleasing design as design that’s more usable.
- von Restorff Effect
  - When multiple similar objects are present, the one that differs from the rest is most likely to be remembered.
- Doherty Threshold
  - Productivity soars when a computer and its users interact at a pace (<400 ms) that ensures that neither has to wait on the other
- Find more here: https://lawsofux.com/

Common Patterns

In Web/Mobile UI

UI Patterns

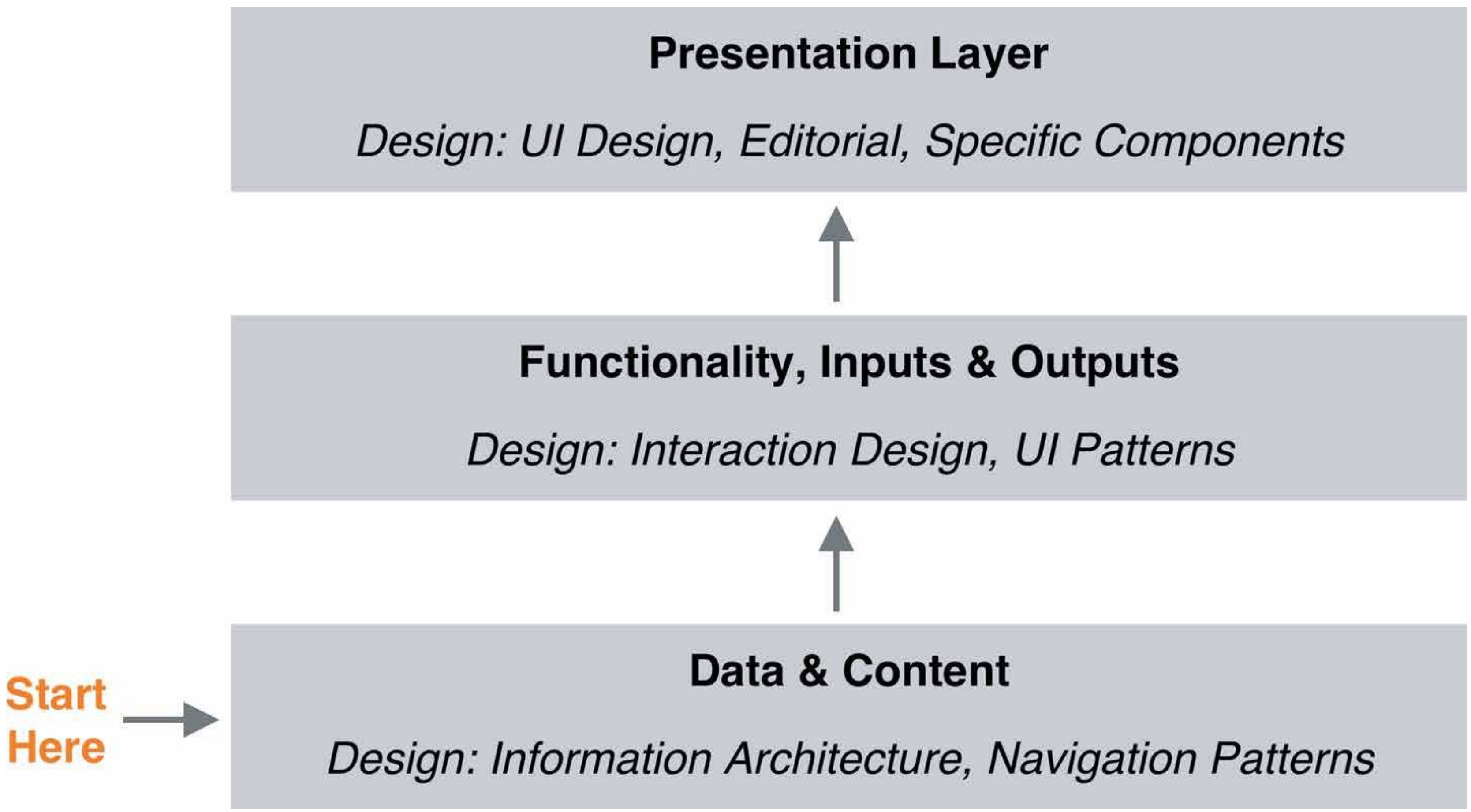


- Contains more than 100 different UI patterns addressing issues of:
  - organizing content
  - navigation
  - layout
  - mobile interfaces
  - listing things
  - doing things
  - collecting data from users
- Provides quick reference to established UI patterns

Organizing Content

- Information Architecture (IA) is the art of organizing and labeling an information space for optimal understanding and use.
- Specifically, IA is using your understanding of your users to design the following:
  - Structures or categories for organizing your content and functionality
  - Different ways that people can navigate through the experience
  - Intuitive workflows or multistep processes for getting tasks done
  - Labels and language for communicating about this content
  - Searching, browsing, and filtering tools to help them find what they're looking for
  - A standardized system of screen types, templates, or layouts so that information is presented consistently and for maximum usability

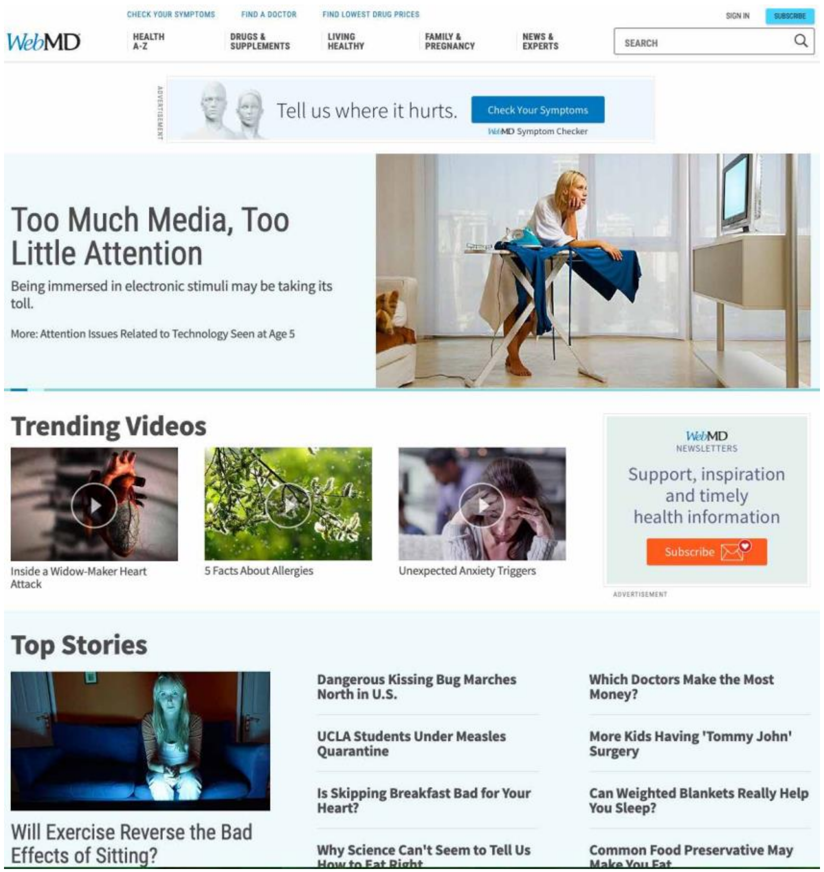
Layered Information Architecture



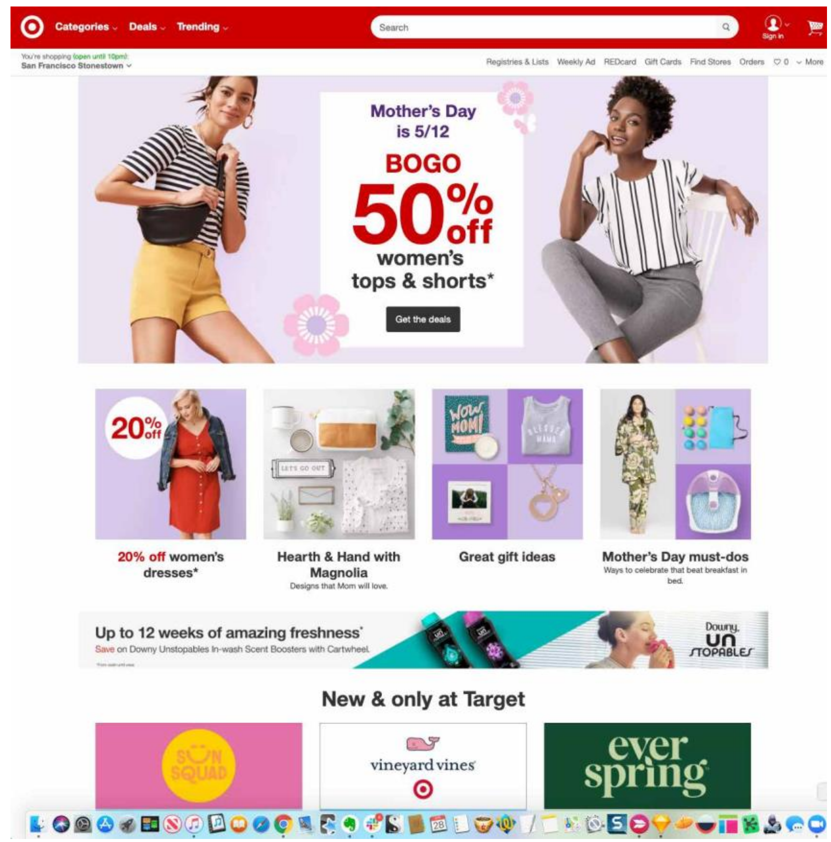
Patterns of IA

- **Feature, Search, Browse**
- A three-element combination on the main page of the site or app:
  - a featured item, article, or product;
  - a search box (expanded by default, or collapsed);
  - a list of items or categories that can be browsed.

Feature, Search, Browse



Content-Centric

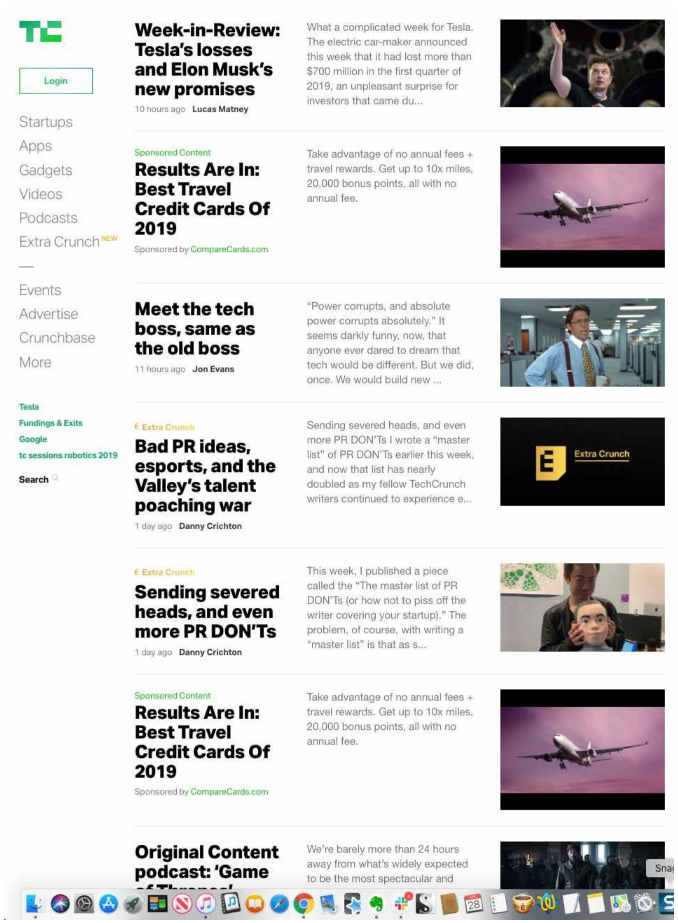


Product-Centric

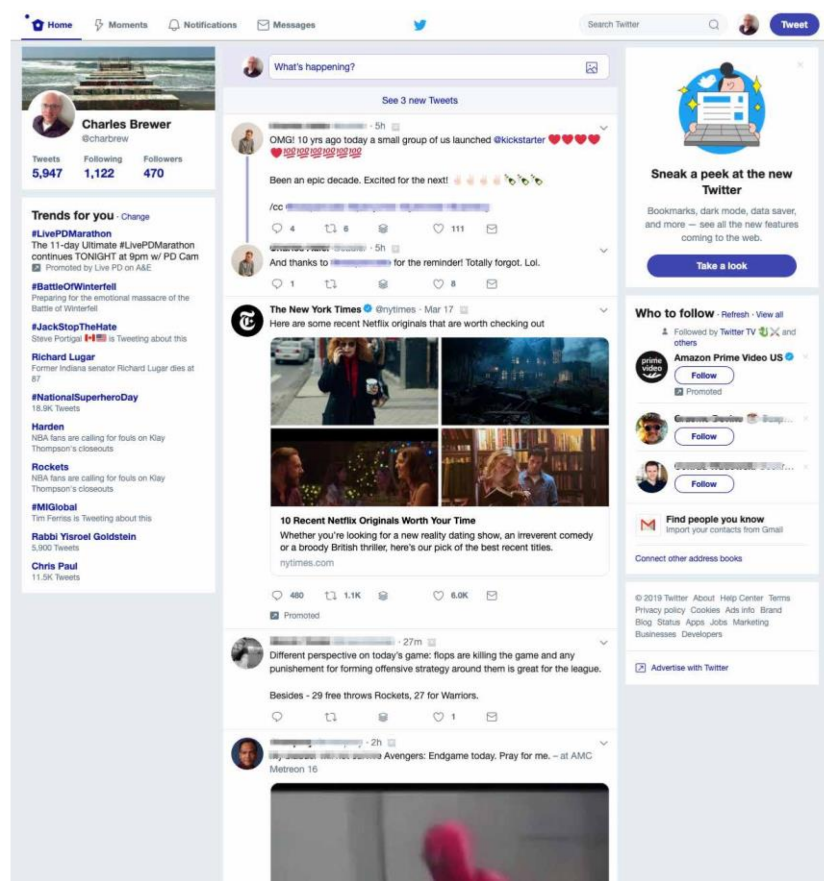
Patterns of IA

- **Streams and Feeds**
- A continuously updated series of images, news stories, web articles, comments, or other content presented in a scrollable, vertical (sometimes horizontal) strip or ribbon.
- The items in the feed are usually presented as "cards," with an image from the article, a headline, introductory copy, and the name of the source with a link.

Streams and Feeds



News/content Stream

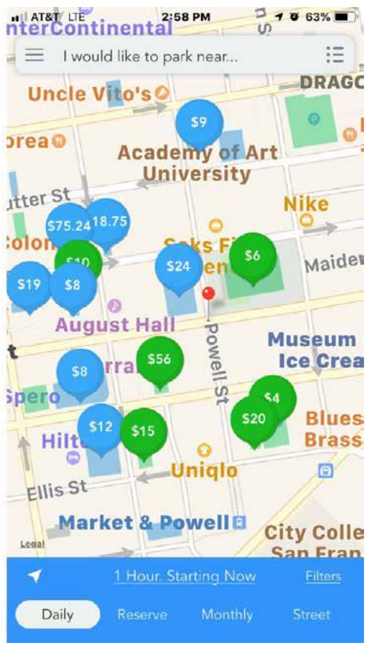


Social Stream

Patterns of IA

- **Mobile Direct Access**
- The first screen presents actionable information without requiring any input or action from the user.
- The app makes assumptions about any settings or queries (such as location or time) related to its primary function, and presents the output for immediate response.
- Making assumptions about what the user is most likely to do also drives what appears on launch.

Mobile Direct Access



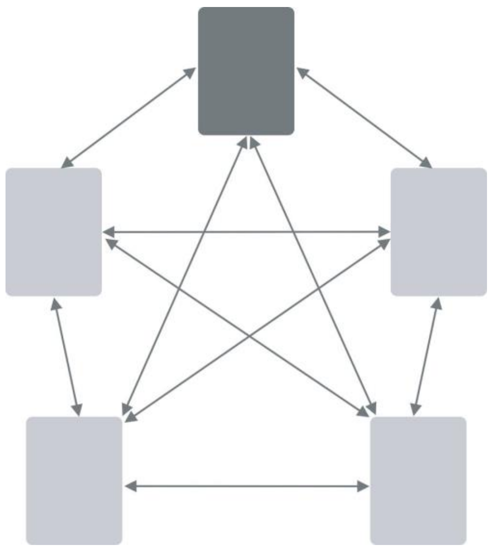
Snap and ParkMe

EventBrit and Weatherbug

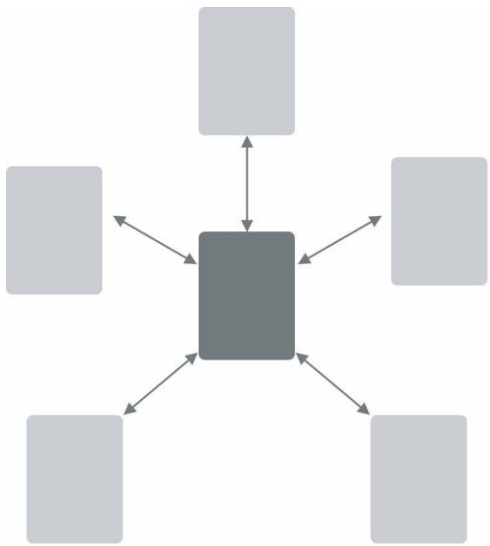
Organizing Navigation

- Navigation helps the user know and understand the information space they are in
- Navigation should answer following questions:
  - Where am I now?
  - Where did I come from?
  - Where should I go next?
- Signposts are features that help users figure out their immediate surroundings
- Wayfinding is what people do as they find their way toward their goal.
  - Unambiguous labels anticipate what you are looking for and instruct where to go
  - Environmental/cultural clues positioned in appropriate locations

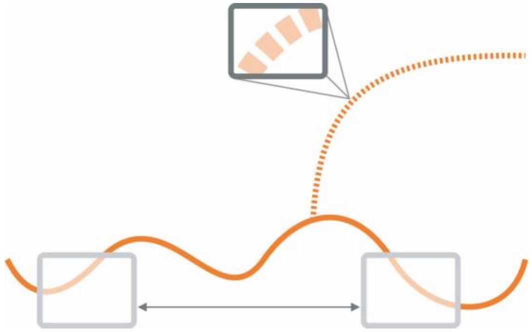
Navigation Models



Connected (e.g. website with global navbar)

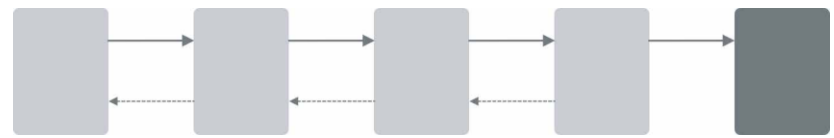


Hub (e.g. iPhone home screen)

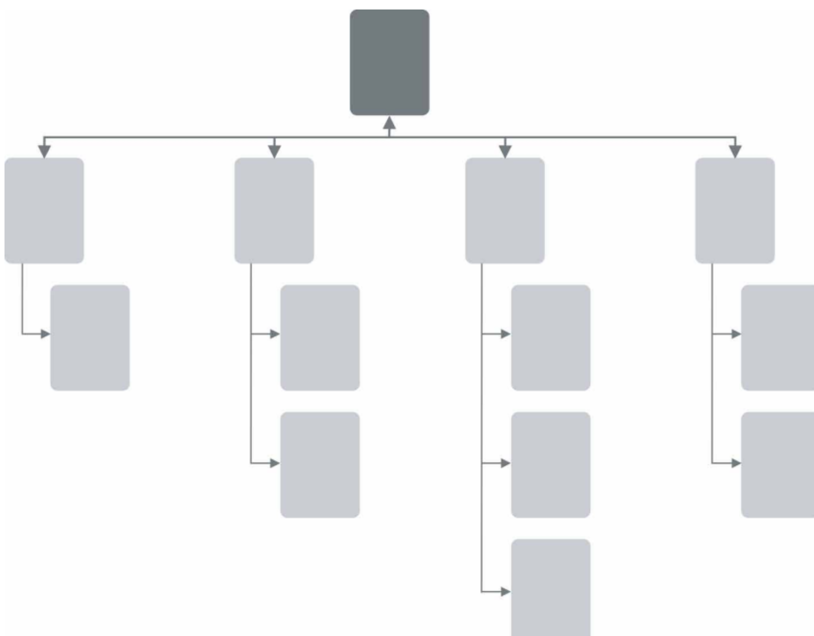


Pan and Zoom (e.g. Google Maps)

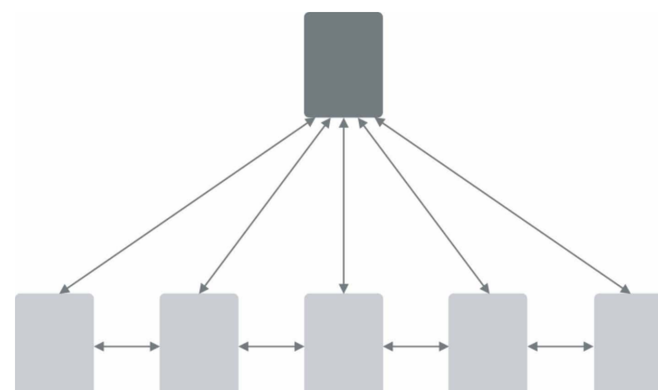
Navigation Models



Step by Step (e.g. wizards)



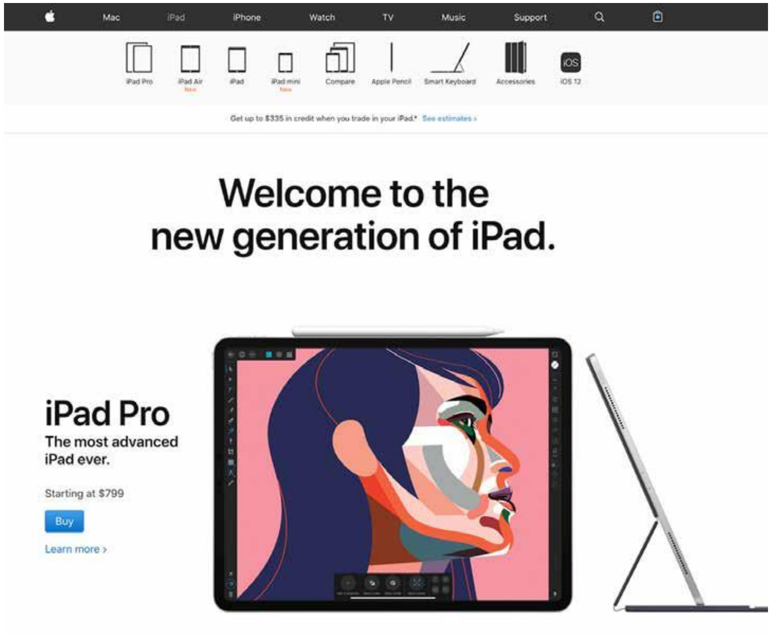
Tree (e.g. olx.uz)

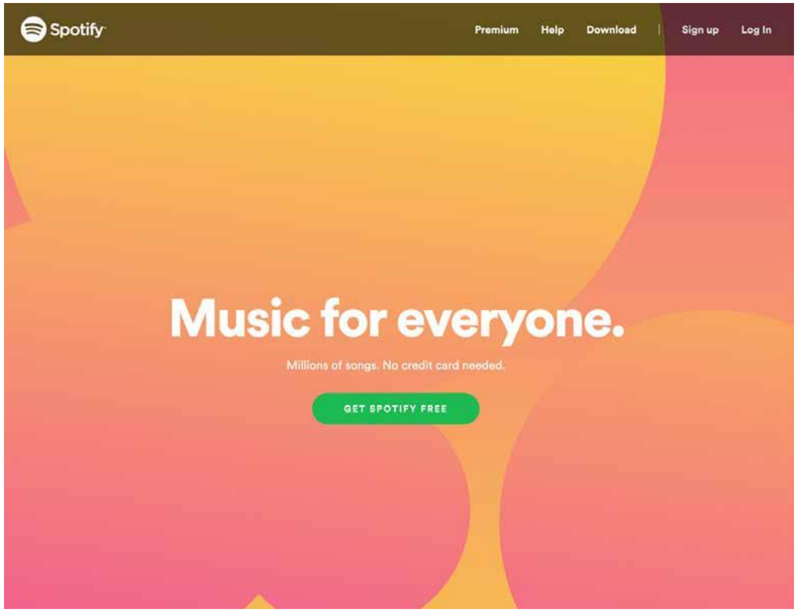


Pyramid (e.g. photo galleries)

Patterns of Navigation

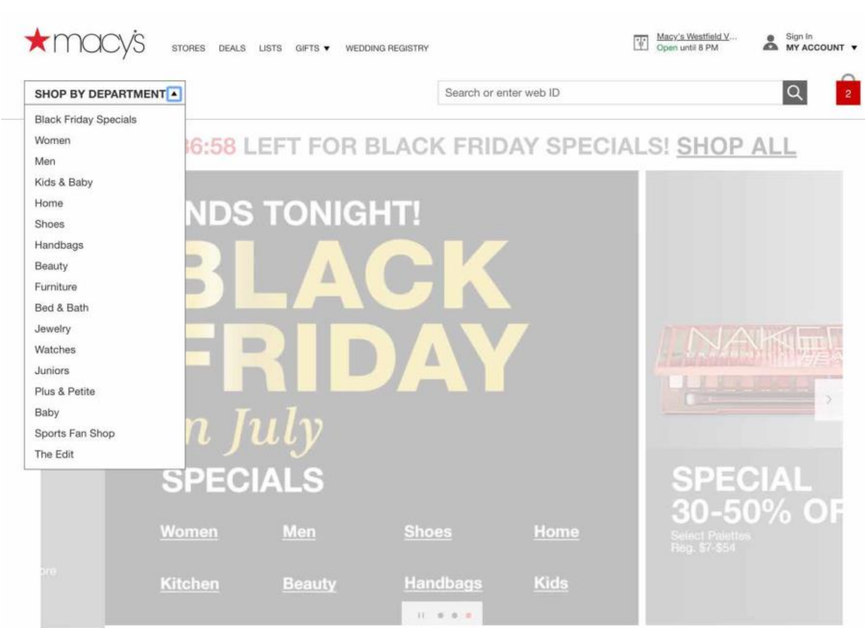
- Clear Entry Points
  - Present only a few main entry points into the interface so that the user knows where to start.

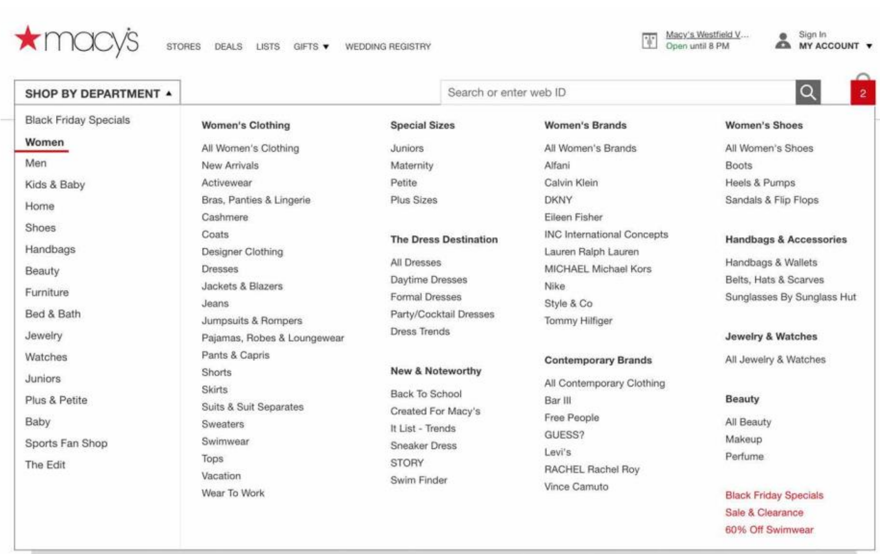




Patterns of Navigation

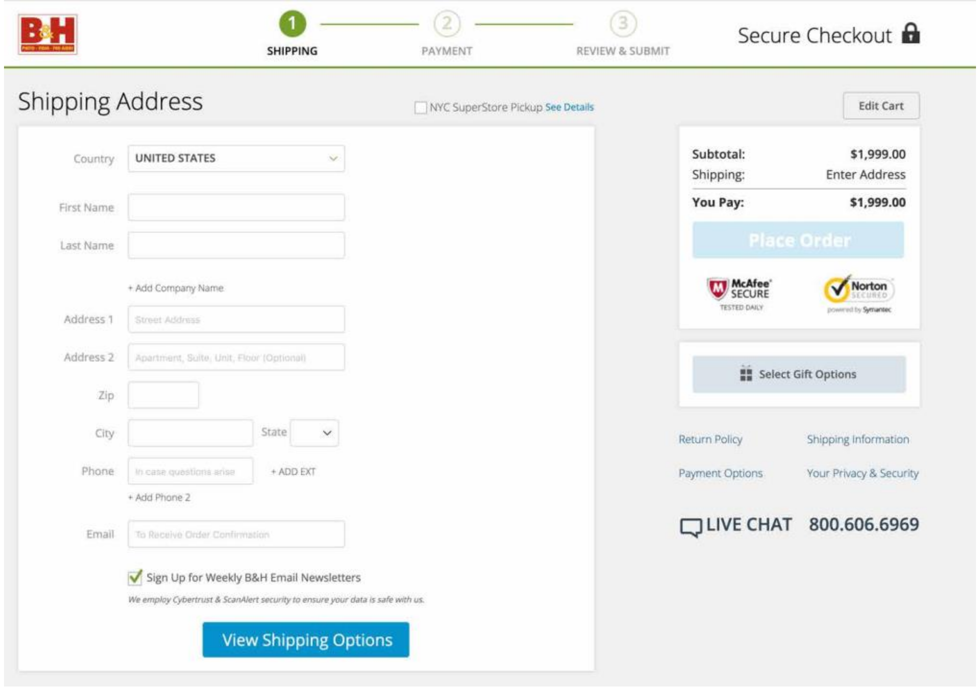
- Fat Menus
  - Display a long list of navigation options in drop-down or fly-out menus.





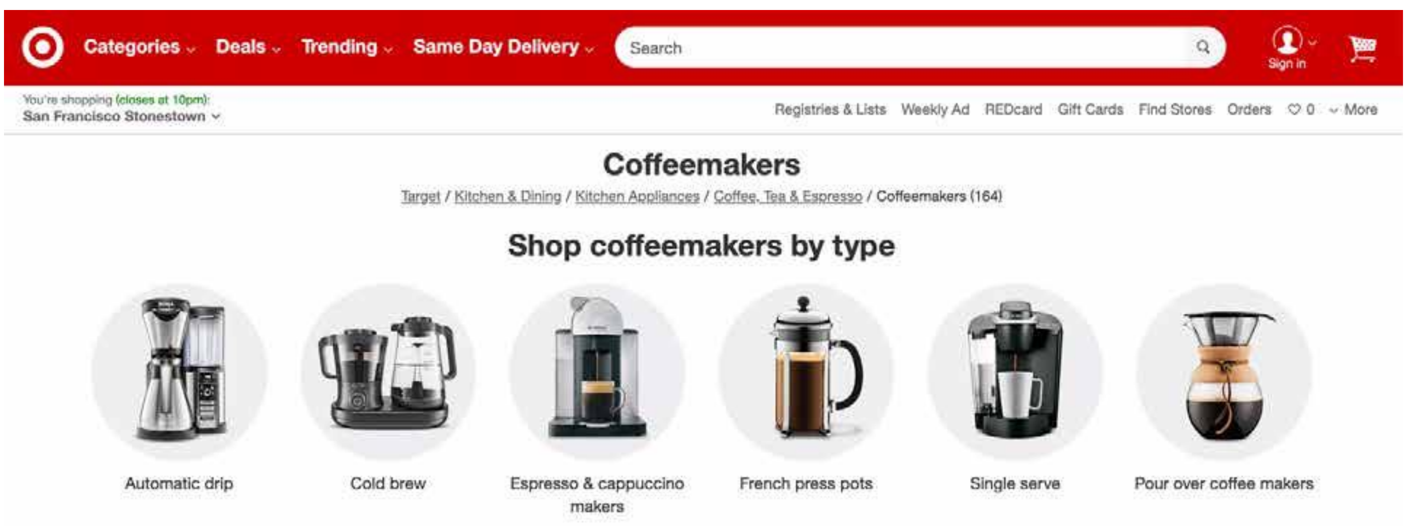
Patterns of Navigation

- Progress Indicator
  - On each page in a sequence, show a map of all the pages in order to show steps in a process, including a “You are here” indicator.



Patterns of Navigation

- Breadcrumbs
  - Shows the path from the starting screen down through the navigational hierarchy, the content architecture of the site, to the selected screen.

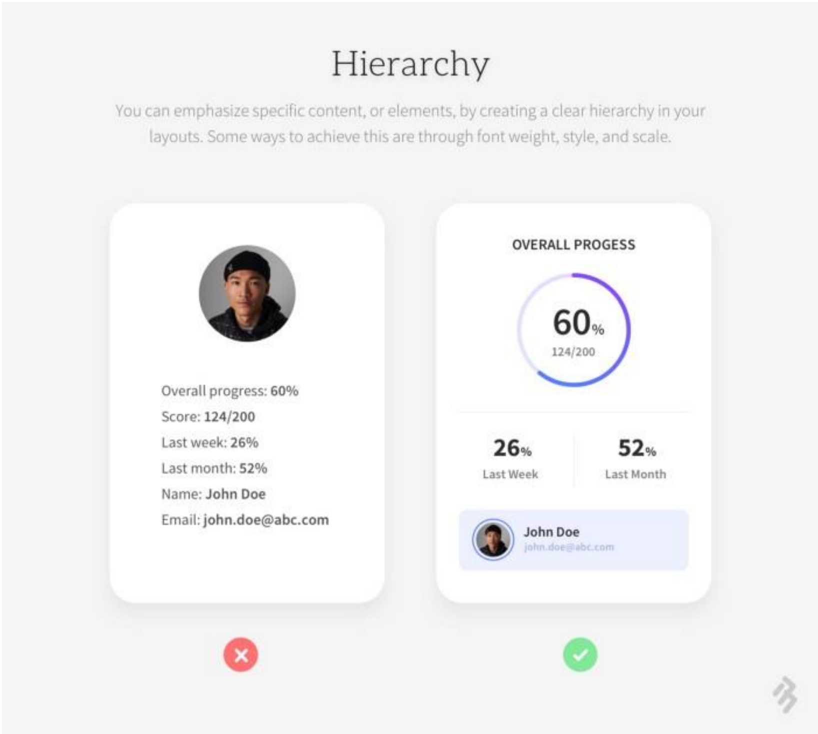


Layouts

- Layout is defined as the particular way informational, functional, framing and decorative elements are arranged in the screen.
- Thoughtful placement of these elements helps to guide and inform your users about the relative importance of the information and functions.

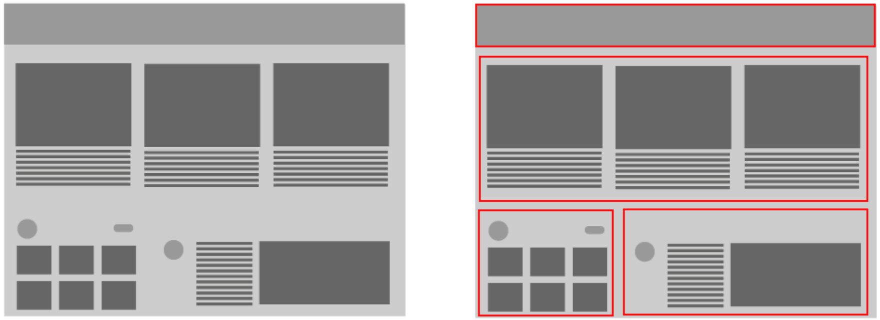
Visual Hierarchy

- Establish visual relationships among layout elements
- Then add different visual weights to each of the elements
- Finally create patterns of eye movement
- Hierarchy can be empathised using following attributes:
  - Size
  - Colour
  - Contrast
  - Density
  - White space



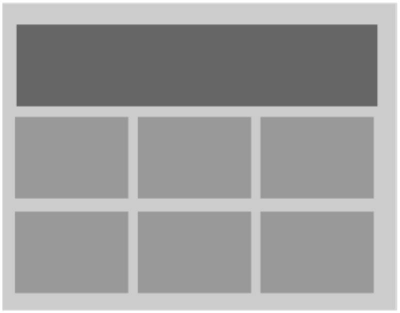
Gestalt Principles - Proximity

- When you put things close together, viewers associate them with one another.
- Grouped items look related.
- Conversely, isolation implies a distinction.



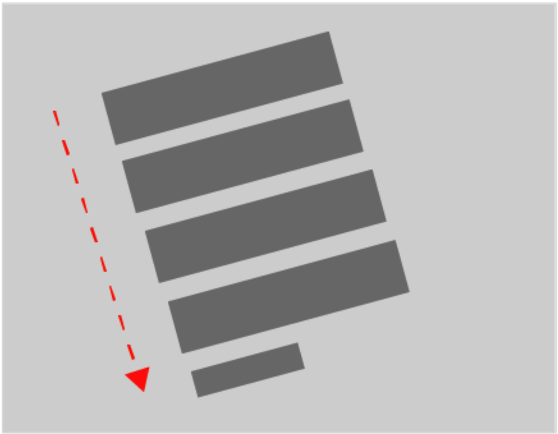
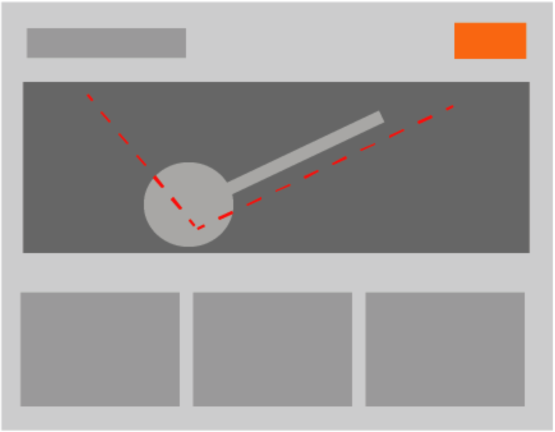
Gestalt Principles - Similarity

- Items that are similar in shape, size, or color are perceived as related to one another.
- If you have a few things “of a type” and you want viewers to see them as equally interesting alternatives, give them an identical graphic treatment.



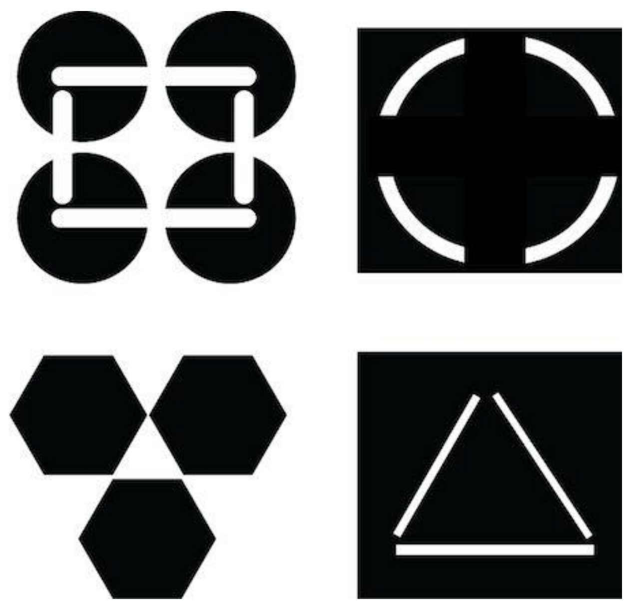
Gestalt Principles - Continuity

- Our eyes will naturally follow the perceived lines and curves formed by the alignment of other elements

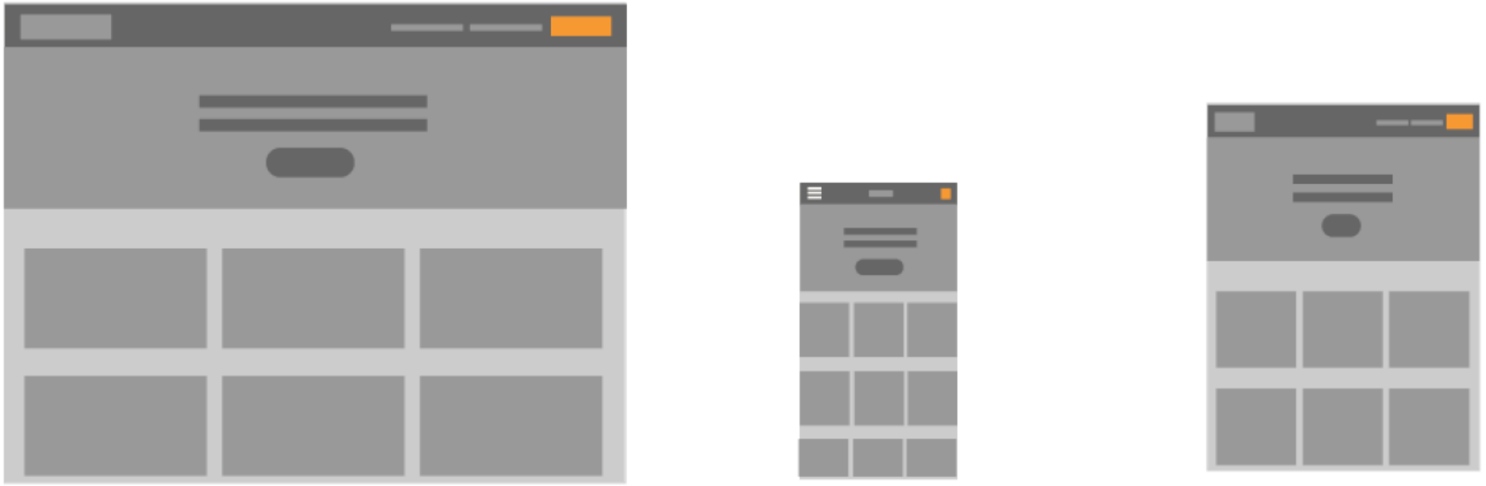
Gestalt Principles - Closure

- The brain will naturally “close” lines to create simple closed shapes such as rectangles and blobs of whitespace, even if they aren’t explicitly drawn for us.



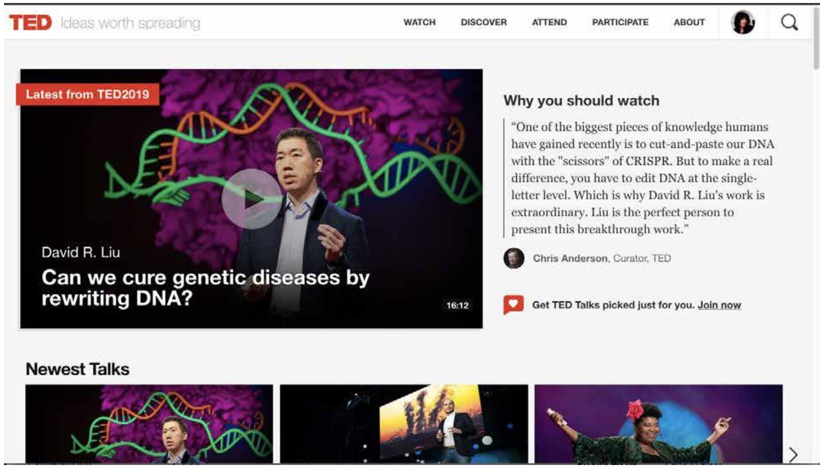
Layout Patterns

- **Grid of Equals**
- Arrange content items, such as search results, into a grid or matrix.



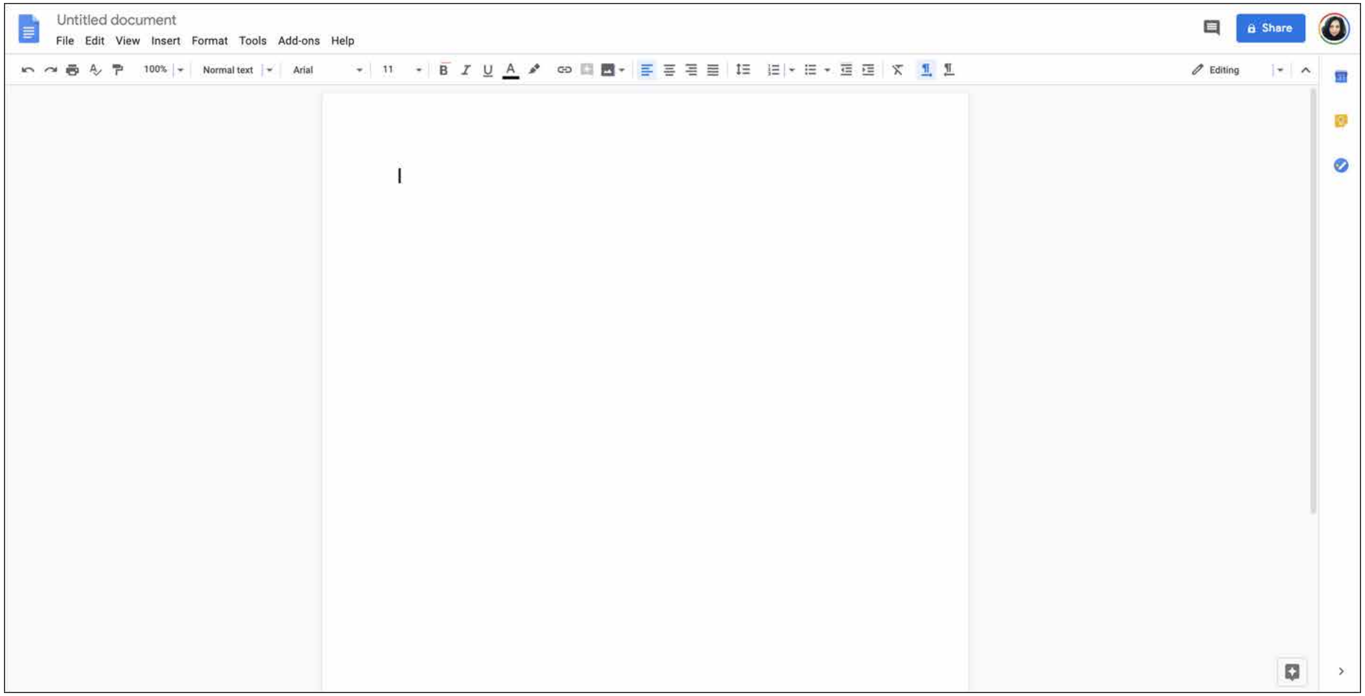
Layout Patterns

- **Visual Framework**
- Across an entire app or site, all screen templates share common characteristics to maintain a consistent layout and style.



Layout Patterns

- **Center Stage**
- The task at hand is placed front and center at most times in the user experience.



And Many More Patterns in the Book…

- Mobile Interfaces
  - Vertical Stack
  - Filmstrip
  - Touch Tools
  - Bottom Navigation
  - Collections and Cards
  - Generous Borders
- Doing Things
  - Button Groups
  - Hover Tools
  - Smart Menu Items
  - Preview
  - Cancellability
  - Prominent "Done"
- Getting User Input
  - Forgiving Format
  - Structured Format
  - Fill-in-the-Blanks
  - Input Hints
  - Password Strength Meter
  - Autocompletion
- Showing Complex Data
  - Datatips
  - Data Spotlights
  - Dynamic Queries
  - Data Brushing
  - Multi-Y Graphs
  - Small Multiples

Figma